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DualAttlog: Context aware dual attention networks for log-based anomaly detection

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ABSTRACT

Most existing log-driven anomaly detection methods assume that logs are static and unchanged, which is often impractical. To address this, we propose a log anomaly detection model called DualAttlog. This model includes word-level and sequence-level semantic encoding modules, as well as a context-aware dual attention module. Specifically, The word-level semantic encoding module utilizes a self-matching attention mechanism to explore the interactive properties between words in log sequences. By performing word embedding and semantic encoding, it captures the associations and evolution processes between words, extracting local-level semantic information. while The sequence-level semantic encoding module encoding the entire log sequence using a pre-trained model. This extracts global semantic information, capturing overall patterns and trends in the logs. The context-aware dual attention module integrates these two levels of encoding, utilizing contextual information to reduce redundancy and enhance detection accuracy. Experimental results show that the DualAttlog model achieves an F1-Score of over 95% on 7 public datasets. Impressively, it achieves an F1-Score of 82.35% on the Real-Industrial W dataset and 83.54% on the Real-Industrial Q dataset. It outperforms existing baseline techniques on 9 datasets, demonstrating its significant advantages.

1. Introduction

During the past decade, the uninterrupted enlargement of software systems has led to a sharp escalation in complexity, resulting in notable impediments to system maintenance efforts. As widely recognized by both academia and industry, as software scales up and its structure becomes more complex, manually managing and maintaining these systems becomes increasingly difficult (Chen et al., 2019; Chen, Yang, et al., 2020; Chen, Zhang, He, et al., 2020; Jiang et al., 2020; Zhao et al., 2020). Against this backdrop, logs, as automatically generated data sources recording system events and operational status, have become increasingly important.

For large-scale and complex software systems (Decker, Leite, Giommi, & Bonacorsi, 2020; He, Zhu, He, & Lyu, 2016; Yang, Chen, et al., 2021; Zhang et al., 2019), such as widely used online service systems and big data systems that process massive amounts of data, logs have become an indispensable source of information for maintenance personnel. These logs typically exist in the form of semi-structured text, containing rich system status information that is easily parsed by both humans and machines. Through careful analysis of these logs, maintenance personnel can promptly detect abnormal behaviors in the system, thereby rapidly locating and resolving issues (Chu et al.,

2021; Li, Chen, Jing, He, & Yu, 2022; Wang et al., 2020; Yan et al., 2021; Zhi et al., 2019).

Nevertheless, as the system's scale and complexity continue to grow, relying solely on manual inspection of logs to detect abnormalities has become impractical (Chen, Zhang, Li, et al., 2020; He, Zhu, He, Li, & Lyu, 2017; Huang et al., 2020; Vervae, 2021; Xu, Huang, Fox, Patterson, & Jordan, 2009). Consequently, in recent times, automated anomaly detection techniques based on logs have garnered significant attention. These methods utilize data mining and machine learning techniques to extract useful information from massive log data and construct models to automatically detect abnormal behaviors in the system.

For instance, Xu et al. introduced an unsupervised learning technique, utilizing Principal Component Analysis (PCA), for detecting anomalies. They first converted log data into numerical vectors, then used PCA to reduce the dimensionality of these vectors, and ultimately, the existence of anomalies was ascertained by measuring the distance between the diminished data points and the distribution of typical state data. This method effectively extracts key information from complex log data and accurately detects abnormal behaviors in the system.

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Moreover, Lou, Fu, Yang, Xu, and Li (2010) extracted invariance from console logs for anomaly detection. By observing the log output during normal system operation, they extracted a series of invariance rules, like the regular appearance of particular events in a prescribed sequence or the sustenance of a constant rate for designated events. When violations of these invariance rules occur during system execution, it can be considered that the system has undergone an anomaly. This method, based on a deep understanding of system behavior, can precisely detect anomalies that may lead to system failure.

Despite their effectiveness in various cases, existing log-based anomaly detection methods commonly lack the required robustness for practical use. This is because these methods typically rely on known log events and sequences extracted from training data to construct detection models. However, in the event of encountering previously unobserved log events and sequences, the functionality of these methods is often seriously impacted. In real-world production environments, most software systems adopt agile development models, there are frequent software updates, leading to subsequent alterations in log data, which include the appearance of novel log events and sequences. This further increases the challenges faced by existing methods.

To tackle this obstacle, this paper introduces a unique log-based anomaly detection approach named DualAttlog. The essence of this method's advantage is its ability to accomplish accurate and robust anomaly detection within dynamically shifting and noisy log data. Specifically, DualAttlog designs an innovative self-matching network that can deeply explore the interactive features between words in log messages. Consequently, it effectively captures the log word-level semantic details within the evolving log messages. Within DualAttlog, we initially compute the interactional details between each word pair within the input log sequence, subsequently formulating self-matching attention vectors utilizing this information. These vectors not only effectively represent the input log sequence but also allow us to perform word-by-word matching within the log sequence, thereby better adapting to the evolution of log messages. This design enables DualAttlog to more accurately identify abnormal logs that deviate from normal patterns. Additionally, to further enhance the performance of DualAttlog, we successfully integrate pre-trained models into the network design. These pre-trained models, having undergone extensive training on vast text data, are capable of extracting profound semantic information. By combining these models with our self-matching network, DualAttlog can capture sequence-level semantic information at the log sequence level, further improving the accuracy of its anomaly detection. Most importantly, DualAttlog successfully connects local semantic information with global semantic information through an innovative context-aware dual attention network design. This design not only maintains excellent discriminative ability but also effectively controls potential information redundancy, enabling DualAttlog to achieve high accuracy while maintaining robustness.

Briefly put, DualAttlog introduces a groundbreaking and efficient method for identifying anomalies in log data. By deeply exploring the dual advantages of word-level semantic information and sequence-level semantic information in log data, DualAttlog achieves accurate and robust anomaly detection in constantly changing and noisy log data, providing strong support for software system maintenance efforts.

The significant contributions of this paper are outlined as follows:

(1) Undertaking the exploration of a novel semantic encoding technique for log sequences: In the semantic encoding, of log sequence, we propose an innovative semantic encoding scheme, which includes two modules: log word-level semantic encoding, and log sequence-level semantic encoding. In the log word-level semantic encoding module, we unveil a self-matching attention mechanism designed to accurately distill the word-level semantic essence of log messages, focusing on the intricate word interactions that occur during their progression. In the log sequence-level semantic encoding module, we utilize a pre-trained model to globally encode the entire log sequence, capturing the overall semantic structure of the log sequence.

This semantic encoding approach, combining both micro and macro dimensions, enables DualAttlog to better understand and represent log data, providing a solid foundation for subsequent anomaly detection.

(2) Investigating the dual attention mechanism across different RoBERTa layers: To fully utilize the powerful capabilities of the RoBERTa model, we further explore the impact of applying the dual attention mechanism across different RoBERTa Transformer layers. By comparing different settings of applying dual attention at a single layer, multiple layers, and all layers, we gain a deep understanding of the practical effectiveness of the dual attention mechanism in anomaly detection tasks. This research not only provides an important basis for the design of the DualAttlog model but also offers new ideas for subsequent studies.

(3) Comprehensive evaluation and benchmark testing: With the aim of comprehensively gauging the performance of DualAttlog, we conducted a series of rigorous experiments encompassing seven public datasets along with two industrial datasets from actual deployment scenarios. The results of our experiments underscore the superior performance of DualAttlog over other baseline models across these datasets, while also demonstrating its strong robustness. Even when faced with constantly changing and noisy log data, DualAttlog maintains stable anomaly detection performance. This accomplishment unequivocally showcases the efficacy and dependability of DualAttlog in real-world scenarios.

The subsequent parts of this paper are structured in the following way: Section 2 gives an overview of the techniques that are connected to our work; Section 3 presents a comprehensive account of our proposed log-based anomaly detection model, DualAttlog, highlighting its structure, operating principles, and defining characteristics; Section 4 describes the specific implementation process of DualAttlog, including data preprocessing, model training, and optimization; Section 5 experimentally validates the performance of DualAttlog on various datasets and conducts detailed analysis and discussion of the results; Section 6 offers a comprehensive overview of the paper's key research contents and accomplishments, coupled with suggestions for future research trajectories and their anticipated outcomes.

2. Related work

2.1. Rule-driven methods

The utilization of deep learning techniques has introduced novel and unprecedented opportunities in the field of detecting anomalies based on logs. Prior to the widespread adoption of deep learning, Traditional techniques in log-based anomaly detection primarily depended upon rule-guided and association-oriented methods (Jafarian, Masdari, Ghaffari, & Majidzadeh, 2021). While these methods can be effective in specific scenarios, they have significant limitations. Rule-based methods often depend on predefined rules constructed from expert knowledge (Hela, Amel, & Badran, 2018), which requires a significant amount of time and manpower, and they struggle to adapt to the complexity and dynamic changes of log data. Association-based methods, on the other hand, mine association rules from accumulated log data to detect anomalies (Lim, Singh, & Yajnik, 2008). However, they are often inefficient in dealing with voluminous log data and can be easily influenced by noise.

Traditional log parsing methods, such as using regular expressions for system fault diagnosis (Vaarandi, Blumbergs, & Kont, 2018; Zhang, Fan, & Guo, 2018), also face challenges. These methods require manual customization of regular expressions to extract specific log events, which is not only time-consuming but also labor-intensive. Moreover, as software systems evolve and log formats diversify, manually crafted regular expressions often fail to adapt to new changes. When faced with complex software systems, these methods often cannot capture comprehensive patterns to guide anomaly detection. Therefore, to meet practical needs, it is necessary to explore techniques that transcend rules and patterns.

2.2. Machine learning-based methods

Numerous researchers have devoted themselves to employing machine learning techniques in log-based fault detection, with the objective of boosting the accuracy and speed of anomaly detection. For instance, using a Support Vector Machine classifier, [Liang, Zhang, Xiong, and Sahoo \(2007\)](#) were able to successfully harness the powerful classification capabilities of machine learning in their analysis of log data, successfully detecting faults from event logs. [Chen, Zheng, Lloyd, Jordan, and Brewer \(2004\)](#) trained a decision tree model to detect anomalies in application operations, effectively capturing the associative relationships between log events through the hierarchical structure of the decision tree. [Farshchi, Schneider, Weber, and Grundy \(2015\)](#) suggested a regression model aimed at detecting anomalies in cloud systems by analyzing log data, thus extending the reach of machine learning in the field of log anomaly detection.

Moreover, [Breier and Branišová \(2015\)](#) conducted a thorough investigation of various classical supervised classification models applied to log-based anomaly detection, providing valuable references and insights for subsequent researchers. These research attempts not only reveal the vast capabilities of machine learning in identifying log anomalies but also pave a solid path for future research endeavors.

However, traditional machine learning methods, despite their accomplishments in detecting anomalies from logs, still possess notable constraints. Similar to rule-based and correlation-based methods, traditional machine learning methods have certain advantages in real-world scenarios but often lack robustness in handling unstable log data. As a result of the ever-changing and diverse nature of log data, traditional machine learning methods struggle to capture all possible anomaly patterns, resulting in limited performance in practical applications.

Therefore, we must strive to further improve the precision and reliability of anomaly detection utilizing log data, it is necessary to explore more advanced and effective techniques. As a branch of machine learning, deep learning possesses powerful feature extraction and pattern recognition capabilities, introducing fresh possibilities for detecting anomalous behaviors from logs. Leveraging the power of deep learning, we are capable of processing and analyzing complex log data with greater efficiency, automatically learning and extracting meaningful features, ultimately enabling the accurate detection of anomalous behaviors.

2.3. Deep learning-based methods

In the last few years, the swift growth of deep learning has given new momentum to the realm of anomaly detection using logs. [Zhang et al. \(2016\)](#) initiated the application of LSTM models to model log keys, enabling the prediction of anomalous behaviors in log sequences. [Du, Li, Zheng, and Srikumar \(2017\)](#) broadened the application of LSTM models by predicting the next occurrence in the log sequence and comparing it to actual data for anomaly detection. Through the training of stacked LSTM models to simulate both regular and irregular operational log samples related to events, [Vinayakumar, Soman, and Poornachandran \(2017\)](#) successfully improved the accuracy of anomaly detection. [Wang, Zhang, Wang and Cao \(2021\)](#) proposed a logging sequence anomaly detection method based on contrastive adversarial training and dual feature extraction. This method employs BERT and VAE to extract semantic features and statistical features of logging sequences respectively, and combines these two types of features for anomaly detection of logging sequences. At the same time, a novel contrastive adversarial training method is adopted to train the model. [Adeba, Kim, and Kwak \(2024\)](#) presented a log anomaly detection method called SaRLog, which aims to address the challenges of balancing model complexity and generating semantically meaningful representations for downstream detection models. By leveraging the contextual semantic information extraction capabilities of Bidirectional Encoder Representations from Transformers (BERT) and the few-shot

learning abilities of the Siamese network, it overcomes the limitations of existing methods.

However, these methods still have certain limitations when dealing with varying log data. Most of them adopt one-hot encoding vectors to process log events, which often performs poorly when handling newly emerging log events. Additionally, most of these methods rely on supervised learning, and the acquisition of labeled data in real-life scenarios often requires significant resources and domain-specific knowledge and experience, limiting their application scope to some extent.

To overcome these limitations, some researchers have begun to borrow natural language processing (NLP) techniques to analyze log data. As a type of natural language sequence, log data possesses rich contextual and semantic information, providing new insights for NLP-based anomaly detection models. By merging LSTM models with TF-IDF weights, [Zhang et al. \(2016\)](#) accomplished specific results in the prediction of abnormal log messages. By leveraging WordVec and traditional classifiers like SVM and random forests, [Bertero, Roy, Sauvanaud, and Trédan \(2017\)](#) detected anomalies in log events, further confirming the efficacy of NLP techniques in anomaly detection within logs.

Regardless of the progress achieved in utilizing NLP tools for the purpose of processing log data, these methods still have some shortcomings. Firstly, most of these models focus solely on the granularity of individual log events, ignoring the overall nature of log sequences. In practical applications, log events are often not isolated but interconnected and form sequences. Therefore, focusing solely on individual log events may not fully capture the contextual information of anomalous behaviors.

Secondly, unlike traditional NLP models that emphasize contextual and semantic information, these anomaly detection models typically overlook semantic information when processing log data. Although log data differs from natural language text, they contain rich semantic information crucial for accurately identifying anomalous behaviors. Consequently, the crucial aspect of enhancing the performance of log-based anomaly detection lies in effectively integrating semantic information into the model.

To cope with the issues stated above, we propose a model named DualAttlog for log anomaly detection, which leverages deep learning and NLP techniques. This model not only considers word-level semantic information within log sequences but also takes into account sequence-level semantic information. Additionally, it leverages a context-aware dual attention network to seamlessly combine word-level semantic information and sequence-level semantic information in log data. By doing so, DualAttlog can more accurately identify anomalous behaviors, strengthening the precision and robustness of anomaly detection.

3. Our proposed approach

3.1. Overview of the proposed model

We will explore the nuances of our model in this segment, and [Fig. 1](#) outlines its structural design.

Initially, the raw logs are processed through two branches. The initial semi-structured log data is transformed into structured log events by the first branch, utilizing log parsing techniques. Following that, every word within each log event is converted into a corresponding word vector. These word vectors are subsequently processed by a self-matching attention network, which identifies and extracts semantic information specific to each word within the log sequence. This enables the model to focus on important information fragments within the log sequence while ignoring irrelevant information, resulting in a more accurate and robust semantic representation vector at the word level.

Skipping the log parsing step, the second branch proceeds to input the raw logs directly into the RoBERTa ([Liu et al., 2019](#)) model. This

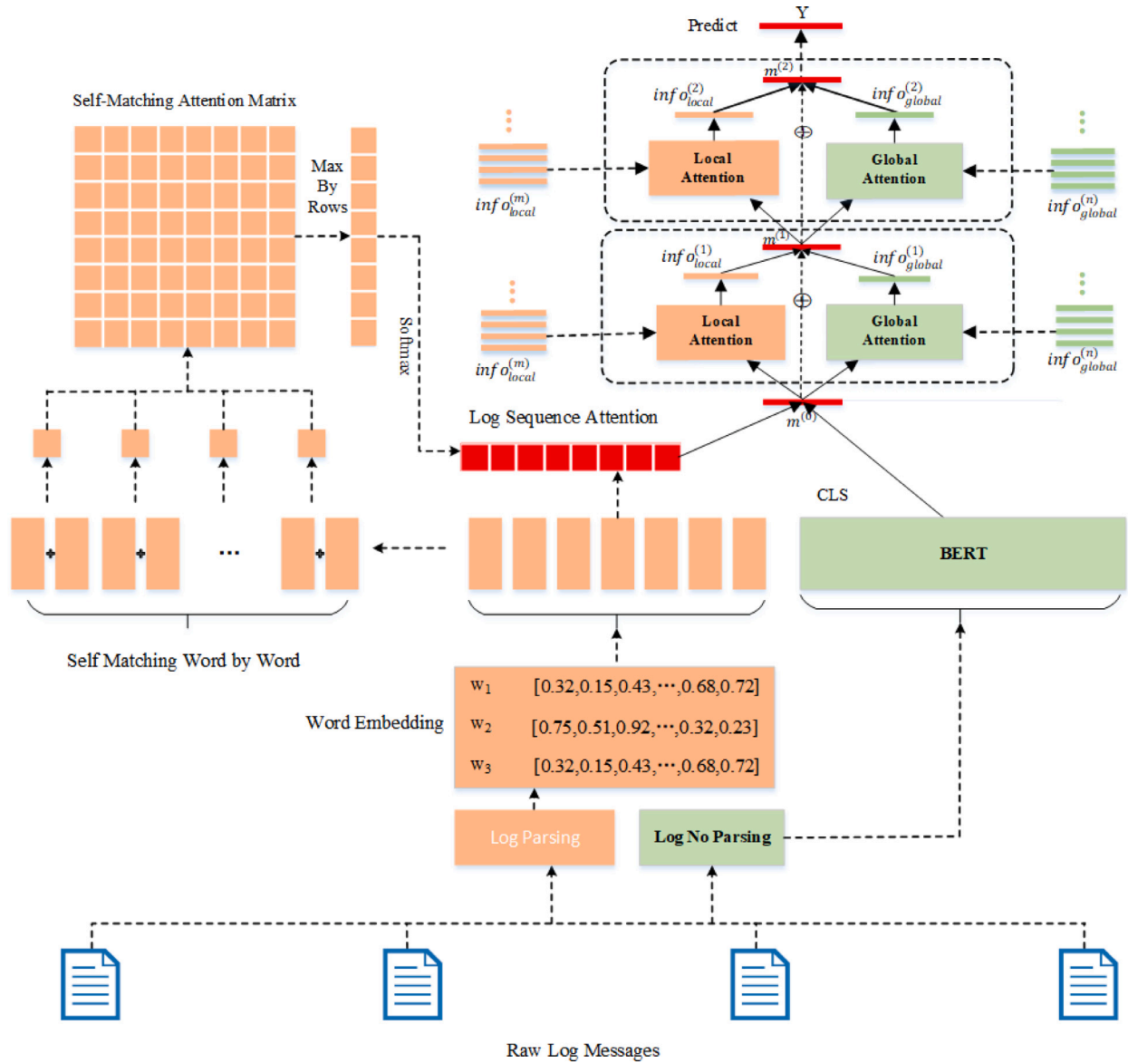


Fig. 1. Model overall architecture.

model then transforms the entire log sequence into a semantic vector with a fixed dimension, focusing on the log sequence level.

Finally, to further integrate these vectorized log sequences, we utilize a context-aware dual attention network to combine the information from both branches. This allows us to better fuse the log information, as a result, the performance of the anomaly detection task based on logs is greatly enhanced.

3.2. Word-level semantic module

In this section, we elaborate on the word-level semantic encoding of log sequences. In the detection of log anomalies, encoding the semantic information at the word level within log sequences is of utmost importance. Word-level semantic information pertains to the manner in which words within log messages interact and the significance they carry within a given context. A self-matching attention network is leveraged to acquire the semantic information at the word level within log sequences. The self-matching attention network is an attention mechanism that can automatically learn and highlight important parts of the log sequence while ignoring irrelevant noise information (Yang, Zhao, et al., 2021). Specifically, To begin with, we extract inter-word

interaction features from the logs that have been parsed. Then, we fuse the word-specific details with the word-level semantic encoding of the log sequence, facilitated by the self-matching network (Yang, Zhao, et al., 2021). Once the preceding steps are completed, we determine a constant number of vectors representing word-word interactions throughout the entire log sequence.

For the log sequence ($S = [s^1, s^2, \dots, s^g]$), the self-matching attention network creates a self-attention-based feature vector: ($f_a = S \cdot a$), where ($a \in R^n$). According to the study in Yang, Zhao, et al. (2021), semantic inconsistency can be employed as a predictive factor for anomaly detection utilizing log data. Consequently, the self-attention vector (a) serves to detect inconsistencies that arise during the evolution of the log sequence.

Xiong, Zhang, Zhu, and Yang (2019) pointed out that incoherence in sentences is mainly caused by conflicts between words, and they delved deeper into the interactions occurring between each word-word pairing. Drawing upon this, they proposed an 'intra-attention' network, which computes the inner product of each word pair to represent their interaction information. Taking cue from the previous idea, we adapt this concept to log sequences and articulate the interaction information ($w_{(i,j)}$) among words (i) and (j) in a given log sequence as detailed here:

$$w_{i,j} = \mathbf{e}_i \cdot \mathbf{e}_j^T \quad (1)$$

In this context, (e_i) and (e_j) denote the embedding vectors of words occupying the (i) th and (j) th positions in the log sequence.

In our research, we employ the aforementioned technique in a comparable fashion to explore potential deep semantic data by contrasting information among words. However, we have noticed that the use of the inner product operation to calculate joint information tends to significantly dilute the informational distinctiveness between individual words. This may be because the inner product mainly focuses on capturing the similarity between two feature vectors and contributes little to processing log sequence information. Furthermore, word embeddings are designed to capture word occurrence patterns, but they may not necessarily be beneficial for our study of log sequences. Consequently, the inner product operation may not fully capture all the conflicts within log sequences, posing a significant challenge for our analysis.

To mitigate this restriction, we propose a parameter matrix that builds on the principles of the co-attention method. The co-attention network proposed by Lu, Yang, Batra, and Parikh (2016) has provided us with valuable insights, especially in addressing visual question answering tasks (Lu et al., 2016). They incorporate a correlation matrix to integrate the feature map G of the input image and the representation T of the textual question, thereby constructing a more comprehensive and insightful interaction model. The formula for calculating the correlation matrix M is as follows:

$$M = \tanh(T \cdot W_a \cdot G) \quad (2)$$

Here, W_a represents the attention weight matrix, which plays a crucial role in the model and helps us better capture the relationships between different input elements.

We employ joint activation methods, including row-wise and column-wise maximization, to simultaneously regulate the attention weights associated with (G) and (T) . In order to strengthen the joint information capture between words, we enhance this approach by inserting a weight matrix connecting word pairs. For the specific word pair (e_i, e_j) , the joint feature vector $(w_{i,j})$ is determined in the following manner:

$$w_{i,j} = \tanh(\mathbf{e}_i \cdot \mathbf{M}_{i,j} \cdot \mathbf{e}_j^T) \quad (3)$$

In this context, $(w_{i,j} \in R)$ encodes the joint data between $word(i)$ and $word(j)$, whereas $(\mathbf{M}_{i,j} \in R^{k \times k})$ functions as a parametric matrix.

Incorporating all joint information $(w_{i,j})$ for (i, j) varying from 1 to n , the computation of the self-matching information matrix (W) proceeds as described:

$$\mathbf{W} = \begin{pmatrix} w_{1,1} & w_{1,2} & \dots & w_{1,n} \\ w_{2,1} & w_{2,2} & \dots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n,1} & w_{n,2} & \dots & w_{n,n} \end{pmatrix} \quad (4)$$

Employing the maximization activation approach, we determine the self-matching attention vector a by computing the intermediate vector $m \in R^n$ through maximizing the elements of W on a row-by-row basis, as detailed below.

$$m_i = \max(w_{i,1}, w_{i,2}, \dots, w_{i,n}), \forall i \in (1, 2, \dots, n) \quad (5)$$

In conclusion, we generate a semantic information representation at the word level for log messages by constructing interactive features among words, fusing self-matching attention networks, and portraying them as vectors with a fixed number of dimensions. This representation method can effectively capture key information and local semantic relationships in log messages, providing strong support for subsequent anomaly detection tasks.

3.3. Sequence-level semantic module

In this section, we will detail how to use a RoBERTa (Liu et al., 2019) model to encode the overall semantics of log sequences in order to capture sequence-level semantic features within them. RoBERTa (Liu et al., 2019) is an improved version based on BERT (Kenton & Toutanova, 2019), which achieves a profound understanding of textual context information through pre-training on large-scale corpora.

Firstly, we represent a log event sequence of length g as $S = [s^1, s^2, \dots, s^g]$. Each log event s^i contains a series of words or tokens that constitute the specific content of the event.

To input the log event sequence into the BERT model, we need to perform a series of embedding operations. These include token embedding, segment embedding, and position embedding. To introduce the log event sequence into the RoBERTa (Liu et al., 2019) model, we must undertake a set of embedding steps. Specifically, token embedding maps each word or token to a fixed-dimensional vector space; segment embedding distinguishes different log events or sentences; and position embedding provides information about the position of words within the sequence. These embedding operations collectively constitute the input to the RoBERTa (Liu et al., 2019) model.

Following the insertion of the embedded log event sequence into the RoBERTa (Liu et al., 2019) model, we leverage its powerful capabilities to process and analyze log data. The RoBERTa (Liu et al., 2019) model, a deep learning model rooted in Transformers technology, is comprised of several stacked encoder layers, each layer utilizing self-attention mechanisms and feed-forward neural networks.

Firstly, The RoBERTa (Liu et al., 2019) model relies heavily on the self-attention mechanism as one of its fundamental components., allowing it to consider all other words in the sequence when processing each word. This mechanism enables RoBERTa (Liu et al., 2019) to capture global contextual information within log event sequences, namely the complex relationships between each word and other words. For log analysis, this global contextual information is particularly crucial as it helps us understand the contextual environment of log events, thereby enabling more accurate identification of anomalies or patterns.

Secondly, the feed-forward neural network in the RoBERTa (Liu et al., 2019) model further processes the features output by the self-attention mechanism. Through the feed-forward neural network, the model performs nonlinear transformations on the captured global contextual information, extracting higher-level feature representations. These feature representations play a pivotal role in the subsequent tasks of detecting log anomalies.

The entire log event sequence's semantic representation is denoted by the special token $[cls]$ present in the RoBERTa (Liu et al., 2019) model's output. The vector output corresponding to the $[cls]$ token is utilized as the holistic contextual representation for the complete log event sequence, denoted as f_l , as detailed in the following equation:

$$f_l = RoBERTa_{cls}(S) \quad (6)$$

It is worth noting that this vector incorporates all semantic details from the log event sequence, making it suitable for use in subsequent anomaly detection tasks. Additionally, The RoBERTa (Liu et al., 2019) model's output dimension is standardized, preserving the consistency of the output vector's dimension across different lengths of input log event sequences. This provides a convenient way to handle variable-length log data.

In summary, by utilizing a RoBERTa (Liu et al., 2019) model as an encoder layer, we can effectively capture global semantic features from log data and represent them as fixed-dimension vectors. These vectors have the potential to act as input features for future anomaly detection models, thereby potentially enhancing the accuracy and efficiency with which anomalies are detected.

3.4. Log context aware dual attention module

Log context aware dual attention module are a crucial component of the model proposed in this paper, particularly when dealing with sequential data. By enabling the model to concentrate on the most significant sections of the input data, they effectively bolster its ability to express and learn. The attention mechanism is particularly important when processing complex temporal features and long sequences, such as log sequences.

In our model, we introduce two types of attention mechanisms: local semantic attention(word-level semantic information) and global semantic attention(sequence-level semantic information). These mechanisms operate independently, enhancing the model's ability to capture different features.

Local Semantic Attention Mechanism: This mechanism primarily focuses on local, short-term dependencies. When processing log sequences, it helps the model capture frequent short-term patterns, such Word-level semantic information in system states. By assigning greater weights to these short-term patterns, the model can better understand and predict future events.

Global Semantic Attention Mechanism: Unlike the local semantic attention mechanism, the global semantic attention mechanism focuses more on global, long-term information. When dealing with long sequence data, this mechanism aids the model in capturing long-term trends and patterns. For instance, when analyzing system logs, global semantic attention can help the model understand the overall system state and behavior patterns across different time points.

These two attention mechanisms operate independently, capturing different features and patterns. However, by complementing each other, they jointly improve the model's ability to comprehend and forecast complex log sequences. In the following sections, we will elaborate on how these two attention mechanisms function within our model.

3.4.1. Local semantic attention of log sequence

The main objective of local semantic attention is to produce a context vector by giving preferential attention to certain sections of the word-level semantic encoding input from the log sequence. In other words, local semantic attention zeroes in on crucial words within each input sequence, enabling the derivation of a context vector. For step k , the context vector $\log_local^{(k)}$ for the local semantic log sequence is obtained by the following formula:

$$\log_local^{(k)} = Local_Atten(\{\log_local_j\}_{j=1}^N, m_{\log_local}^{(k-1)}) \quad (7)$$

Here, $m_{\log_local}^{(k-1)}$ serves as a memory vector, encoding all the information taken into account until the $k-1$ th step.

Using a two-layer feed-forward neural network (FNN), the attention weights $\{\alpha_{\log_local,n}^{(k)}\}_{n=1}^N$ are computed, and then a softmax function is applied to them. This entire process is outlined in the following manner:

$$h_{\log_local,j}^{(k)} = \tanh(W_{\log_local}^{(k)} \cdot \log_local_j) \odot \tanh(W_{\log_local,m}^{(k)} \cdot m_{\log_local}^{(k-1)}) \quad (8)$$

$$\alpha_{\log_local,j}^{(k)} = \text{softmax}(W_{\log_local,h}^{(k)} \cdot h_{\log_local,j}^{(k)}) \quad (9)$$

$$\log_local^{(k)} = \sum_j \alpha_{\log_local,j}^{(k)} \cdot \log_local_j \quad (10)$$

where $W_{\log_local}^{(k)}$, $W_{\log_local,m}^{(k)}$, and $W_{\log_local,h}^{(k)}$ are network parameters, $h_{\log_local,j}^{(k)}$ is the hidden state, and $\odot$ denotes element-wise multiplication.

3.4.2. Global semantic attention of log sequence

The global semantic attention mechanism first computes a context vector $\log_global^{(k)}$ by weighted summing each log sequence in the

input sequence $\{\log_global_i\}_{i=1}^T$. The weights are calculated based on the memory vector $m_{\log_global}^{(k-1)}$ from the previous step, which encodes the accumulated information from previous steps of the model. This is represented as:

$$\log_global^{(k)} = Global_Atten(\{\log_global_i\}_{i=1}^T, m_{\log_global}^{(k-1)}) \quad (11)$$

where $\log_global^{(k)}$ is the context vector and $m_{\log_global}^{(k-1)}$ is the memory vector.

Specifically, for each sentence $\log_global_i$ in the input sequence, the model first computes a hidden state $h_{\log_global,i}^{(k)}$ through a nonlinear transformation (such as the tanh function). This hidden state is a function of the log sequence itself and the memory vector, combining them through element-wise multiplication.

This process can be expressed by the following equations:

$$h_{\log_global,i}^{(k)} = \tanh(W_{\log_global}^{(k)} \log_global_i) \odot \tanh(W_{\log_global,m}^{(k)} m_u^{(k-1)}) \quad (12)$$

$$\alpha_{\log_global,i}^{(k)} = \text{softmax}(W_{\log_global,h}^{(k)} h_{\log_global,i}^{(k)}) \quad (13)$$

$$\log_global^{(k)} = \sum_i \alpha_{\log_global,i}^{(k)} \log_global_i \quad (14)$$

Next, the model uses another network layer to compute the attention weights $\alpha_{\log_global,i}^{(k)}$. Typically, this procedure encompasses subjecting the hidden states to a linear transformation and subsequently normalizing them using the softmax function, guaranteeing that the collective sum of all weights equals unity. These weights reflect the model's focus on each log sequence in the input sequence at the current step k .

Finally, utilizing the attention weights computed, the model performs a weighted aggregation of the input sequence, thereby obtaining the global semantic context vector $\log_global^{(k)}$. This vector integrates information from different log sequences in the input sequence and emphasizes those most relevant to the current task. This context vector will be used as input for subsequent steps, guiding the model for further processing and prediction.

Unlike focusing solely on local information, the global semantic attention mechanism boasts the advantage of encompassing global information within the input sequence. By doing so, the model gains a deeper and more holistic comprehension of the overall structure and content within the log sequences, thereby more accurately identifying key sentences and features relevant to the current task. This is especially important for processing long sequence data since long sequences often contain rich global information and long-term dependencies.

3.5. Model fusion mechanism

The schematic representation of our dual attention network is presented in Fig. 1. At the heart of our network lies the fusion memory connection vector, which is introduced in this section. Alternatively stated, DualAttlog functions through the transmission of memory connection vectors, functioning as a conduit that combines local semantic details with global semantic insights extracted from log sequences. It also plays a role in gradually transmitting information. According to the following equations, $m^{(k)}$ serves as a fusion memory vector, responsible for storing the accumulated local and global semantic details obtained from earlier steps k :

$$m^{(k)} = m^{(k-1)} + \log_local^{(k)} \odot \log_global^{(k)} \quad (15)$$

Here, $\log_local^{(k)}$ represents the local semantic vector, while $\log_global^{(k)}$ signifies the global semantic vector, both derived from the preceding section.

By means of this fusion technique, both local and global attention are simultaneously steered, i.e., $m^{(k)} = m^{(\log_local^k)} = m^{(\log_global^k)}$. Consequently, the two attention mechanisms can function in close cooperation.

Table 1
Detailed statistics of public datasets.

Dataset name	Dataset size	Number of log messages	Number of abnormal logs
HDFS (Zhu et al., 2019)	1.58GB	11,175,629	362,793
BGL (Zhu et al., 2019)	1.207GB	4,747,963	949,024
BGP (Zhu et al., 2019)	1.04GB	11,428,282	1,276,742
Thunderbird (Zhu et al., 2019)	27.367GB	211,212,192	43,087,287
Spirit (Oliner & Stearley, 2007)	30.289GB	272,298,969	78,360,273
Liberty (Oliner & Stearley, 2007)	29.5GB	266,991,013	191,839,098
Zookeeper (Zhu et al., 2019)	10.4M	74,380	49,124

The global context vectors, $r^{(0)}$ and $u^{(0)}$, form the basis for the initial memory vector, $m^{(0)}$:

$$m^{(0)} = \log_local^{(0)} \odot \log_global^{(0)} \quad (16)$$

$$\log_local^{(0)} = \tanh \left(P^{(0)} \frac{1}{N} \sum_j t_j \right) \quad (17)$$

$$\log_global^{(0)} = \frac{1}{T} \sum_i u_i \quad (18)$$

During this process, dual attention and memory are updated k times. Finally, We predict the detection results by means of a single-layer softmax classifier, which incorporates cross-entropy loss, and the input for this classifier is the memory unit $m^{(K)}$:

$$p_i = \text{softmax}(W_f \cdot m^{(K)} + b) \quad (19)$$

Here, $p_i \in R^2$ indicates whether the input log sequence is regular or irregular, and $W_f \in R^{2 \times c}$ along with $b \in R^2$ are the adjustable parameters.

3.6. Training objective

For the task of log-based anomaly detection, the loss function adopted corresponds to the conventional cross-entropy formulation. Its mathematical expression is given below.

$$J(\theta) = - \sum_{i=1}^N [y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)] + \lambda \cdot R \quad (20)$$

Here, N stands for the total number of instances in the training dataset, and y_i indicates the true label assigned to the i th log sequence. The set of model parameters is encompassed by θ , including $\{M_{i,j}, U, V, g, W_f, b\}$. The regularization term, expressed as $R = \|\theta\|_{L_2}$, measures the magnitude of the model parameters, and λ is a hyperparameter that controls the impact of the regularization term.

4. Experimental setup

4.1. Dataset

The public datasets used in this article include seven types, specifically: HDFS dataset (Zhu et al., 2019), BGL dataset (Zhu et al., 2019), BGP dataset (Zhu et al., 2019), Thunderbird dataset (Zhu et al., 2019), Spirit dataset (Oliner & Stearley, 2007), Liberty dataset (Oliner & Stearley, 2007), and ZooKeeper dataset (Zhu et al., 2019). Detailed information is provided in Table 1.

In addition to the public datasets, we have also collected two authentic industrial log datasets from large enterprises, namely Real-Industrial W dataset and Real-Industrial Q dataset. The detailed information is presented in Table 2. Specifically, Real-Industrial W is a financial transaction service based on a microservices architecture, serving millions of users. Real-Industrial Q is a WEB-based distributed online commodity trading service, also serving millions of users. These two services generate millions of log sequences daily. Both are composed of multiple components and are deployed with weekly updates.

Table 2
Detailed statistics of production environment datasets.

Dataset name	Dataset size	Number of log messages	Number of abnormal logs
Real-Industrial W	1.36 GB	3,275,859	214,780
Real-Industrial Q	1.15 GB	2,973,597	185,620

The introduction of new log events during system updates poses significant challenges for anomaly detection based on log data mining, making them highly suitable for validating our approach. We have collected log data from dates before and after system updates, ensuring that the new log events in these two datasets were introduced by the deployed updates.

4.2. Training and hyper-parameters

This section focuses on explaining the process of determining fixed hyperparameters for our model. During the pre-training phase, we employed a basic version of RoBERTa to generate pre-trained embeddings. Empirical evidence indicates that a dual attention step count (denoted as K) of 2 yields optimal performance. For optimization, we decided to utilize the Adam Optimizer algorithm, specifying a value of 0.9 for the first momentum coefficient β_1 and 0.999 for the second momentum coefficient β_2 .

Furthermore, We have taken great care in setting the initial learning rate and the L2 regularization parameter. The initial learning rate was chosen from the range of $[1 \times 10^{-9}, 4 \times 10^{-5}, 1 \times 10^{-7}]$, while the L2 regularization parameters were set within the range of $[1 \times 10^{-6}, 4 \times 10^{-7}, 1 \times 10^{-7}]$. Additionally, we experimented with batch sizes of [64, 128, 256]. These hyperparameters were fine-tuned using development sets and subsequently validated on test sets. We performed our experiments on NVIDIA TESLA V100 GPU hardware, leveraging the PyTorch framework.

5. Experimental results

For a more comprehensive evaluation of the performance of our proposed mode, we have formulated four research inquiries and performed corresponding experiments in order to assess the effectiveness of the model introduced in this research.

RQ1: Does the model outlined in this paper have the potential to achieve a higher level of performance in log anomaly detection compared to the most advanced methods?

For the purpose of evaluating the log anomaly detection capabilities of our model, we plan to benchmark it against seven established models across seven publicly available datasets. By adopting this comparison method, we can conduct a thorough evaluation of the proposed model's performance and gain insights into its performance variations across various datasets.

RQ2: What is the contribution of the key model components in the model proposed in this paper to the overall performance?

To answer the above question, we divide the proposed model into three key components (using only local semantic representation, using only global semantic representation, and removing the dual attention network). We conducted experiments on multiple datasets for each of these key components to evaluate their individual contributions to the overall model.

RQ3: How does the size of parameters, particularly the hidden state dimension of the dual attention network and the number of dual attention network steps K , influence the overall model's performance in this paper?

There are many key parameters in the model proposed in this chapter. We selected the iteration steps K of the dual attention network

Table 3
Detailed experimental results on seven public datasets.

Datasets	Metric	DeepLog★	LogAnomaly★	PLELog★★	OC4Seq	LogRobust	NeuralLog	CAT	DualAttlog
HDFS	Precision	88.39%	79.43%	87.85%	89.38%	94.37%	93.65%	94.53%	96.58%
	Recall	83.32%	98.15%	89.52%	89.74%	96.82%	91.47%	97.75%	98.78%
	F1-score	85.78%	87.80%	88.68%	89.56%	95.58%	92.55%	96.11%	97.67%
BGL	Precision	87.26%	81.39%	88.21%	90.53%	95.52%	94.39%	95.68%	97.25%
	Recall	86.34%	98.59%	91.67%	90.21%	97.38%	92.45%	98.84%	98.89%
	F1-score	86.80%	89.17%	89.91%	90.37%	96.44%	93.41%	97.23%	98.06%
BGP	Precision	78.65%	79.82%	83.28%	86.59%	93.62%	91.52%	95.32%	96.45%
	Recall	87.69%	88.91%	89.76%	90.83%	95.57%	92.62%	91.38%	95.65%
	F1-score	82.92%	84.12%	86.40%	88.66%	94.58%	92.07%	93.31%	96.05%
Thunderbird	Precision	86.43%	83.84%	85.37%	86.29%	93.85%	91.58%	93.87%	95.48%
	Recall	85.95%	86.83%	87.29%	88.63%	93.59%	92.38%	90.28%	96.92%
	F1-score	86.19%	85.31%	86.32%	87.44%	93.72%	91.98%	92.04%	96.19%
Spirit	Precision	74.65%	75.86%	76.92%	78.93%	91.85%	89.29%	92.37%	93.52%
	Recall	82.17%	83.62%	84.71%	86.65%	93.24%	93.53%	94.53%	96.56%
	F1-score	78.23%	79.55%	80.63%	82.61%	92.54%	91.36%	93.44%	95.02%
Liberty	Precision	72.13%	73.86%	78.62%	81.53%	92.25%	91.45%	93.32%	94.86%
	Recall	81.57%	87.19%	88.67%	90.12%	94.72%	91.59%	93.84%	95.69%
	F1-score	76.56%	79.97%	83.34%	85.61%	93.47%	91.52%	93.58%	95.27%
ZooKeeper	Precision	75.34%	77.68%	81.25%	82.53%	93.52%	92.95%	95.32%	96.29%
	Recall	82.56%	85.29%	87.62%	90.12%	92.82%	91.59%	96.45%	97.64%
	F1-score	78.78%	81.31%	84.31%	86.16%	93.17%	92.26%	95.88%	96.96%

and the hidden state dimension. We performed experiments utilizing various datasets in order to assess the effect of these key parameters on the overall model's functionality.

RQ4: How do the various layers of RoBERTa, which are utilized in the model introduced in this paper, contribute to its overall performance?

The model introduced in this paper employs the base version of RoBERTa, consisting of a total of 12 layers. We question whether the semantic information extracted from all 12 layers has an identical influence on the proposed model's functionality. To answer this question, We undertook a series of experiments to validate the contribution of diverse layers to the model proposed in this chapter.

5.1. Model comparisons

In addressing research question (RQ1), we conducted a comprehensive evaluation of the performance of DualAttlog in the task of log anomaly detection. To ensure the broad applicability and reliability of our experimental results, we carefully selected 7 publicly available datasets and further introduced 2 datasets from real-world production environments for model training and testing. In the following, we will elaborate on the specific application of these datasets in the evaluation process.

5.1.1. Experiments on public datasets

To ensure the fairness and accuracy of the comparative experiments, this paper carefully selected seven baseline methods and conducted detailed experiments on seven publicly available and unified datasets. For supervised learning methods, including OC4Seq (Wang, Chen, et al., 2021), LogRobust (Zhang et al., 2019), NeuralLog (Le & Zhang, 2021), CAT (Zhang et al., 2022), and DualAttlog, we adhered to strict principles for dataset splitting. Specifically, we divided the datasets into training and testing sets in chronological order, with 80% of the data used for model training and the remaining 20% reserved for evaluating model performance. During the training process, we fully utilized labels as feedback to guide the optimization of the models. However, during the testing phase, we followed the conventional approach of supervised learning models and did not use any form of feedback. Furthermore, to ensure reproducibility and consistency of the experiments, we maintained the same parameter settings as in the original papers during the training process. For unsupervised models (such as DeepLog (Du et al., 2017) and LogAnomaly (Meng et al., 2019)) and

semi-supervised models (such as PLELog (Yang, Chen, et al., 2021)), we also adhered to the original paper's dataset splitting, evaluation methods, and training parameter settings. This approach aims to ensure the objectivity and fairness of the experimental results, enabling a more accurate assessment of the performance of different methods in the field of log analysis.

The evaluation results in Table 3 (where ★ represents unsupervised methods, ★★ represents semi-supervised methods, and the others are supervised methods) show the highest F1 scores highlighted in bold:

(1) Compared to supervised learning methods, unsupervised learning methods (DeepLog (Du et al., 2017) and LogAnomaly (Meng et al., 2019)) exhibited significantly lower effectiveness in aspect of F1 scores. This is because they were originally designed to treat all training samples as normal. As a result, when making inferences, they are inclined to categorize any sample without recognizable log templates as abnormal, leading to a situation where recall is high but precision is low, ultimately causing reduced F1 scores.

(2) Semi-supervised learning methods (PLELog (Yang, Chen, et al., 2021)) outperformed unsupervised learning methods (DeepLog (Du et al., 2017) and LogAnomaly (Meng et al., 2019)) in terms of F1 scores. This is primarily due to their ability to leverage partially labeled data. These labeled data provide some guidance to the model, enabling it to more accurately distinguish between normal and abnormal patterns, thereby improving precision. In contrast, unsupervised learning methods lack explicit label guidance, making it difficult to accurately identify abnormal patterns during training, resulting in lower precision. However, the F1 scores of semi-supervised learning methods are still lower than those of supervised learning methods because they can only utilize partially labeled data. This means that compared to fully supervised methods with abundant labeled data for precise model training, their guidance between normal and abnormal instances is relatively limited.

(3) In the task of log anomaly detection, when it comes to performance, supervised learning methods can be more effective than both unsupervised and semi-supervised learning methods. This is primarily attributed to the requirement of labeled datasets in supervised learning, which provide known input-output correspondences. This correspondence enables the model to directly learn the clear boundaries between abnormal and normal patterns during training. Therefore, when faced with new log data, supervised learning models can more accurately identify anomalies. Consequently, most supervised learning methods (OC4Seq (Wang, Chen, et al., 2021), LogRobust (Zhang et al., 2019),

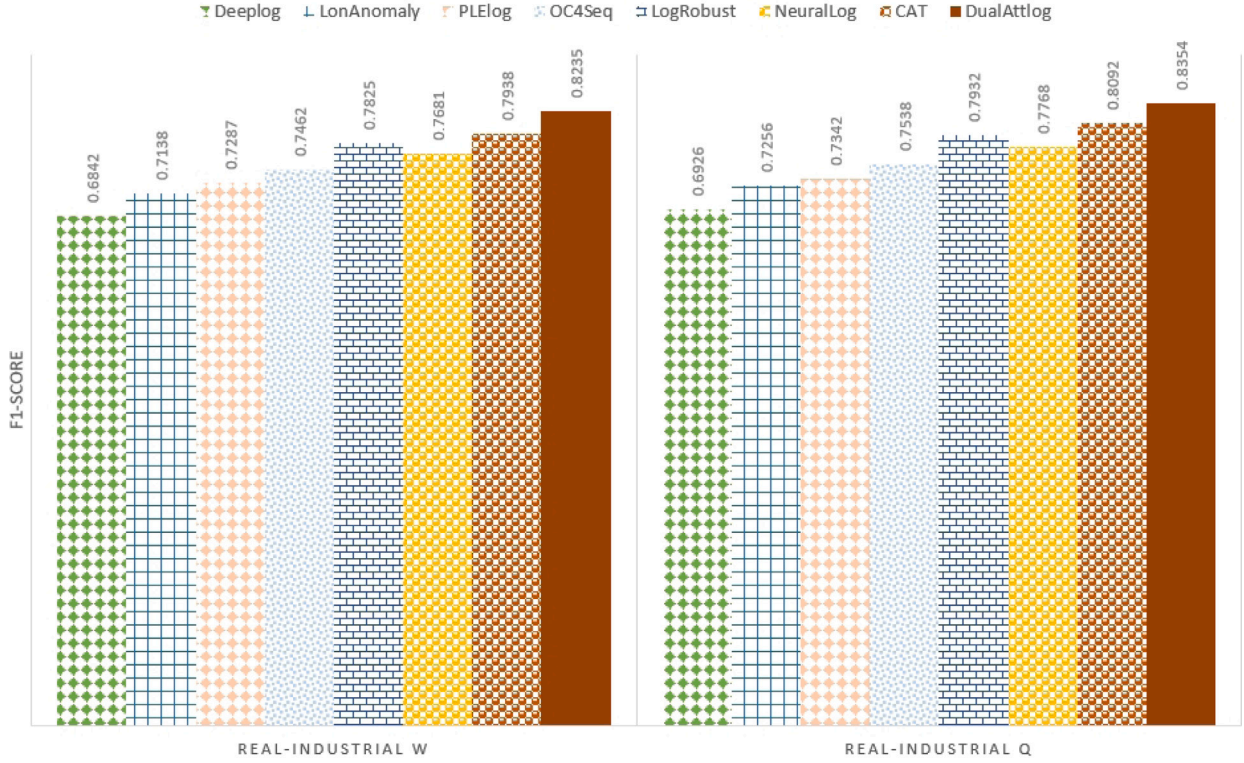


Fig. 2. Detailed results on two production environment datasets.

NeuralLog (Le & Zhang, 2021), and CAT (Zhang et al., 2022)) exhibit good performance.

(4) The method proposed in this paper demonstrates the best performance across all datasets. This is mainly because it takes a holistic view, encompassing both situations where a log parser is required and where it is not. In the case of no log parser being utilized, the method directly utilizes a pre-trained model for global semantic extraction from log sequences (similar to NeuralLog (Le & Zhang, 2021), effectively addressing log noise issues). When a log parser is required, it utilizes a self-matching attention network to extract word-level semantics from logs (similar to LogRobust (Zhang et al., 2019), better handling log evolution issues). Subsequently, by adopting our proposed context aware dual-attention network, it attains the highest level of performance by integrating these two kinds of information.

5.1.2. Experiments on two real-world production datasets

In order to more effectively assess the performance of the proposed model in actual production settings, we collected two datasets from real-world operational systems and divided them in chronological order. Among these, 80% of the data was used as the training set, while the remaining 20% was reserved for testing. During the training process, for all four models (LogRobust (Zhang et al., 2019), NeuralLog (Le & Zhang, 2021), CAT (Zhang et al., 2022), and DualAttlog), we integrated labels as feedback to ensure that the models could learn accurate data patterns.

However, during the testing phase, we strictly adhered to the standard methodology of supervised learning models, refraining from using any form of feedback. This approach aimed to guarantee the objectivity and fairness of the testing results, reflecting the models' performance on unseen data accurately.

For the three selected baseline methods (LogRobust (Zhang et al., 2019), NeuralLog (Le & Zhang, 2021), and CAT (Zhang et al., 2022)), we paid particular attention to maintaining their training parameters consistent with the settings in the original papers. This was done

to ensure the fairness and accuracy of the comparative experiments, allowing our proposed method to be compared with these baseline methods under identical conditions. Fig. 2 depicts the results:

The information in Fig. 2 reveals that, LogRobust (Zhang et al., 2019) and CAT (Zhang et al., 2022) are the most powerful baseline models for log-based anomaly detection tasks. Our DualAttlog model exhibited superior performance on both datasets, as evident from the experimental results. Specifically:

(1) Our model surpassed the benchmark model CAT (Zhang et al., 2022) by 2.97% in F1-Score on the *Real-Industrial W* dataset, achieving significance at $p < 0.05$ on the student t-test. Similarly, it exceeded LogRobust (Zhang et al., 2019) by 2.62% with the same level of significance.

(2) Our model surpassed the benchmark model CAT (Zhang et al., 2022) by 4.10% in F1-Score on the *Real-Industrial Q* dataset, achieving significance at $p < 0.05$ on the student t-test. Similarly, it exceeded LogRobust (Zhang et al., 2019) by 4.22% with the same level of significance.

The reason for our model's superior performance on the two real-world production environment datasets lies in its ability to explore the interactive features between words and capture the semantic relationships among them during the process of log message evolution. Additionally, We calculate the interactive information existing between every individual word and every pair of words within the given log sequence, and utilize a self-matching attention network to handle the log sequence and generate word-level semantic vectors.

Moreover, we utilize the RoBERTa model to encode the entire log sequence and generate a sequence-level semantic vector. Finally, we leverage an innovative context aware dual-attention network for logs to fuse word-level and sequence-level information while reducing the impact of noise on model performance. Consequently, Our final experimental results confirm that our technique attains significant performance when dealing with log-based anomaly detection tasks.

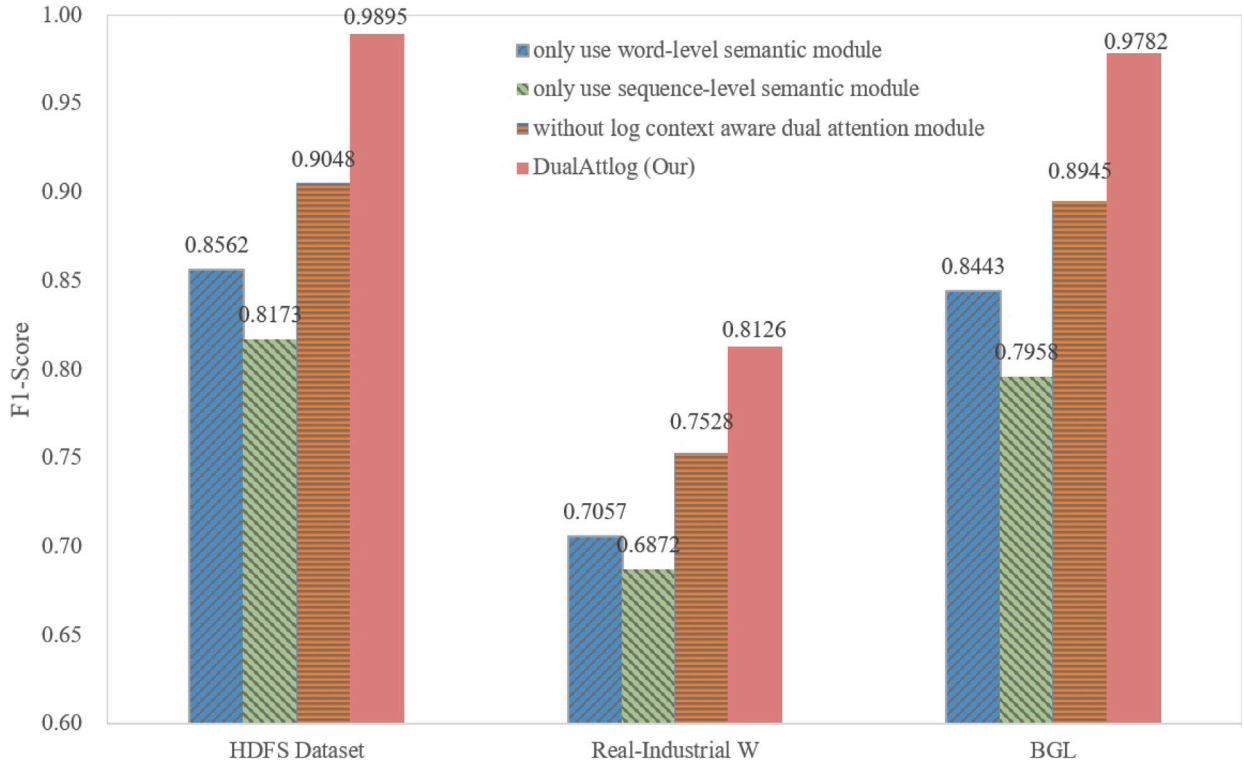


Fig. 3. Ablation study.

5.2. Ablation study

To gain a comprehensive understanding of how each essential part of our model contributes to its overall performance (RQ2), we conducted a set of ablation experiments, decomposing the entire model into three derivative models. To enhance the validation process of our model, we conducted a series of ablation studies across three datasets. The findings of the experiment are depicted in Fig. 3.

The three derivative models are as follows: Model (1): Only use word-level semantic module; Model (2): Only use sequence-level semantic module; Model (3): Without log context aware dual attention module.

From the ablation experiment results shown in Fig. 3, several observations can be made. Firstly, comparing Model (1) and Model (2), Model (2) exhibits a performance decrease of 3.89%, 1.85%, and 4.85% in F1-score on the HDFS, Real Industrial W, and BGL datasets, respectively. This indicates that only use sequence-level semantic module slightly degrades the model's performance compared to only use word-level semantic module. This suggests that, while both are important for our method, the Only use word-level semantic module may be more crucial. This is primarily because the Only use word-level semantic module, by exploring word-word interaction features, captures deep semantic information from the evolution of log messages. Therefore, obtaining rich semantic representations is a valuable solution.

Secondly, comparing Model (3) with both Model (2) and Model (1), the results show that Model (3) outperforms both on the HDFS, BGL, and Real Industrial W datasets. These results indicate that even when the log context aware dual attention module is removed, its performance is still better than using either representation alone. The combined use of both types of information better addresses log-based anomaly detection tasks.

Lastly, we compared Model (3) with our proposed model, DualAttlog. On the HDFS, Real Industrial W, and BGL datasets, Model (3) exhibited accuracy reductions of 8.47%, 5.98%, and 8.37%, respectively. The findings confirm that the log context aware dual attention module plays a crucial role in our proposed method. This primary

reason is that the log context aware dual attention module excels at capturing the semantic relationship between word-level and sequence-level meanings in log messages. Therefore, removing this component can degrade the overall model performance.

In summary, our proposed model utilizes self-matching attention to capture word-level semantic information and RoBERTa to capture sequence-level semantic information. By adopting the log context aware dual attention module, it fully captures interactive features in the sequence at both the word-level and global sequence-level, maximizing log-based anomaly detection performance.

5.3. Sensitivity analysis of parameters

In this part, we assess the effect of several key parameters on the proposed model (RQ3). For instance, the influence of the hidden layer's dimensionality and the number of iterations K within the context aware dual attention network on three datasets (HDFS dataset, BGL, and Real Industrial W).

Initially, we aim to investigate the impact of the hidden layer dimension within the context-aware dual attention network model on its overall performance, we conducted a series of experiments and conducted a detailed analysis of the results. Fig. 4 exhibits the experimental results, revealing several crucial trends.

(1) On three different datasets, the F1 score of the model demonstrates a clear upward trend when the hidden layer dimension falls below 300. This suggests that, within a certain range, increasing the hidden layer dimension can indeed enhance the effectiveness of the model. This stems from the fact that an increased dimension enables the model to gather a richer set of feature information, thereby better understanding the intrinsic structure of the data.

(2) Notably, when the hidden layer dimension is exactly 300, the F1 score of the model reaches its peak. This finding indicates that at this specific dimension, the model can most effectively utilize additional log data feature information to achieve optimal performance. This may be due to the fact that a 300-dimensional hidden layer can provide a feature representation space that is neither too complex nor too

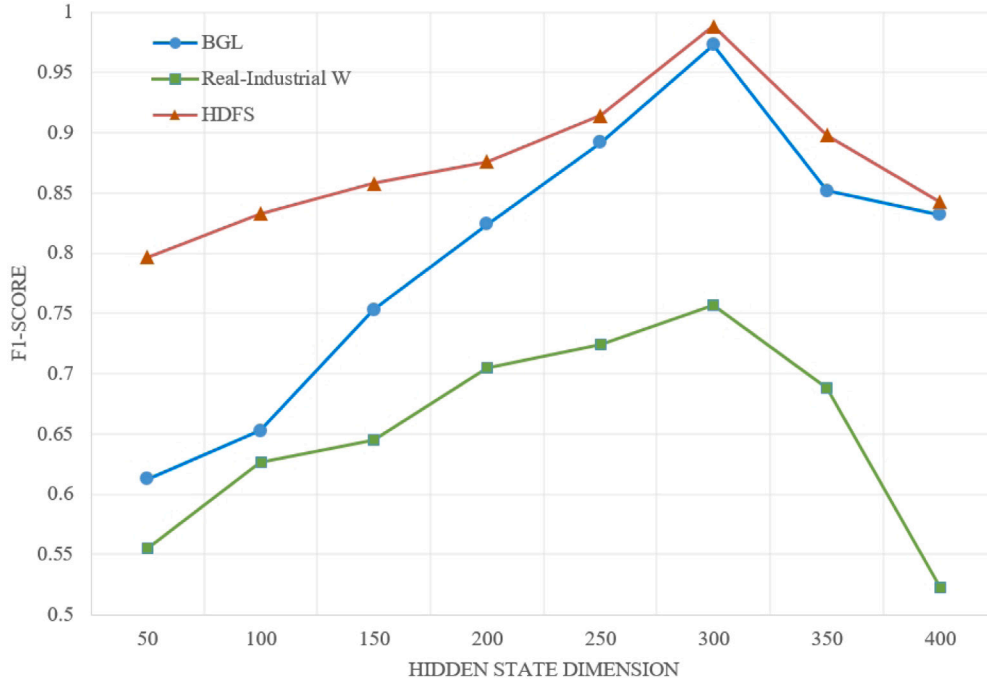


Fig. 4. Hidden layer parameter analysis of context aware dual attention.

simple, enabling the model to fully learn the details of the data while maintaining generalization ability.

(3) When the hidden layer dimension exceeds 300, we observe a decline in the model's accuracy on the validation and test sets. This could be due to the emergence of overfitting. When the model's complexity is too high, it may pay too much attention to noise and details in the training data, ignoring the overall structure of the data. This results in poor performance on unseen data, indicating a decrease in generalization ability. Additionally, insufficient training data may also contribute to the decline in performance. With limited training data, an overly complex model may have difficulty fully learning the inherent patterns of the data, affecting its performance on the validation and test sets.

Overall, our experimental findings reveal that the dimension of the hidden layer exerts a notable influence on the effectiveness of the context-aware dual attention network model. Increasing the dimension within a certain range can improve model performance, but an excessively large dimension may lead to overfitting and a decrease in performance. Therefore, in practical applications, we need to select an appropriate hidden layer dimension derived from the characteristics of the specific task and dataset to gain optimal model performance.

Next, we delve deeper into the impact the steps K in the context aware dual attention network on model performance and conduct a comparative analysis across three different datasets. As shown in Fig. 5, when the steps K in the dual attention network is set to 2, our model reaches the utmost accuracy on all three datasets. This discovery provides important clues for understanding the working mechanism of the model and optimizing its performance.

(1) When steps K is small, the model may face the problem of insufficient fitting. This is because a smaller number of steps means that the model may not fully capture the dependencies between log events, unable to fully extract and utilize useful information from the data. This constraint diminishes the model's comprehension of the inherent structure and patterns in the log data, thereby affecting its accuracy in making predictions and classifications.

(2) When steps K is large, the model may experience overfitting. A longer number of steps makes the model overly focus on the details and noise in the data during training, ignoring the overall structure and

generalization ability. This results in poor performance on unseen data, indicating a decrease in generalization ability. Additionally, A greater number of steps can also result in an increase in the computational demands and training duration required by the model, reducing its efficiency.

(3) Notably, selecting the appropriate steps K is crucial for the performance of the dual attention network. In this experiment, we found that the model performs best when K is set to 2. This may be because, under this setting, the model can fully capture the dependencies between log events while avoiding overfitting and increasing computational complexity. This result provides a useful reference for adjusting model parameters in practical applications.

In summary, through a comparative analysis of steps K in the context aware dual attention network, We realized that steps play a crucial role in determining the model's performance. Selecting the appropriate steps can ensure that the model fully learns from the data while avoiding overfitting and increasing computational complexity, thereby achieving optimal model performance.

5.4. Exploration of distinct RoBERTa layers

To explore the impact that different layers of RoBERTa have on the overall functionality of the model(RQ4), in this paper, we performed a sequence of experimental tests by applying the dual-attention network component on top of different layers of RoBERTa. By comparing the performance of single-layer, multi-layer, and full-layer Transformers, we aimed to reveal the differences in feature extraction and semantic understanding capabilities across various layers of RoBERTa.

In terms of experimental design, We designated the k th layer (where k is 1, 3, 6, 9, or 12) of RoBERTa as the exclusive layer for our experimental analysis. For the configuration of multi-layer Transformers, We step by step enhanced the number of layers and tested combinations such as 1,3, 1,3,6, and 1,3,6,9. Finally, we also evaluated the full-layer Transformer, which involved applying the dual-attention network after each layer of RoBERTa for analysis. All experiments were conducted on the Real-Industrial W dataset to ensure consistency and comparability of the results. The outcomes of the experiments are showcased in Table 4.

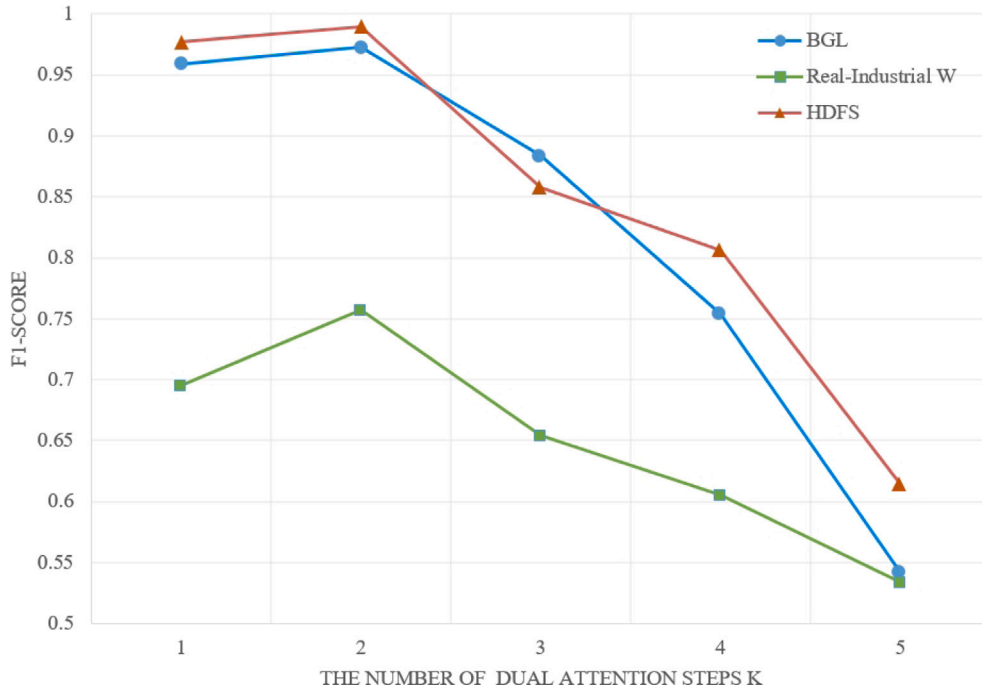


Fig. 5. Steps K analysis of context aware dual attention.

Table 4
Analysis of RoBERTa's layer performance variations.

Layer	single				
	1	3	6	9	12
F1	0.7325	0.7569	0.7197	0.6879	0.6682
Layer	multi			full	
	1,3	1,3,6	1,3,6,9	full	
F1	0.7413	0.7385	0.7294	0.6473	

The findings displayed in Table 4 suggest that shallow layers of RoBERTa tend to yield better performance in the task of log anomaly detection. This observation results from the shallow layers' ability to foster a layered interplay between log features and RoBERTa, enabling the model to more effectively capture key information within log sequences. Furthermore, we also noticed a trend of decreasing performance when applying the context aware dual attention component to various layers of RoBERTa. This phenomenon may be attributed to the overfitting issues arising from multi-layer integration, where the model becomes overly focused on noise and fine details in the training data, thereby reducing its generalization ability on test data.

Based on these findings, we can further optimize the model. For instance, we can experiment with adjusting the weights of different RoBERTa layers or introducing regularization techniques to mitigate overfitting issues. Additionally, investigating the utilization of various pre-trained models or fine-tuning techniques could potentially enhance the model's performance even further.

5.5. Threats to validity

Here are the key threats that have the potential to undermine our effectiveness:

(1) **Subjects:** In this paper, we employed nine distinct datasets: HDFS, BGL, BGP, Thunderbird, Spirit, LibertyZooKeeper, Real-Industrial W, and Real-Industrial Q datasets. HDFS, BGL, BGP, Thunderbird, Spirit and LibertyZooKeeper are open-source datasets, have served as prevalent benchmarks in log-based anomaly detection tasks. Conversely, Real-Industrial W and Real-Industrial Q datasets stem from expansive

real-world software systems. Despite their extensive log counts, the nine datasets have a restricted scope in the quantity of topic systems they cover. They do not cover all domains comprehensively. Moving forward, we aim to expand our dataset repertoire by incorporating more datasets sourced from software systems across diverse domains. This extension in dataset variety will facilitate a more comprehensive evaluation of the efficacy and robustness of the methods proposed in this paper across a broader spectrum of domains.

(2) **Drastic changes:** It is true that our proposed method caters well to typical software maintenance scenarios, such as those in continuous delivery and deployment contexts. However, challenges arise when significant changes occur in the online system's codebase or logging mechanism. For instance, if the original system is replaced, or the new system—while serving similar functions—is developed by a different team and employs a drastically altered logging mechanism, DualAttlog may encounter performance issues if it is not incrementally updated to adapt to these changes. In this kind of situation, where there are major alterations in the system architecture or logging protocols without corresponding updates to DualAttlog, could result in a considerable drop in its performance. Maintaining alignment between the system's evolution and the adaptability of DualAttlog becomes crucial to sustain its effectiveness in anomaly detection under evolving system configurations.

(3) **Noises in labeling:** The meticulous labeling process carried out by engineers for all four datasets in this paper indeed suggests a high degree of accuracy in the dataset annotations. However, it is important to acknowledge the potential introduction of data noise, such as false positives or false negatives, during the phase of manual tagging. Despite the presumption that the noise is minimal in its magnitude, if present at all, it remains a possibility that warrants further investigation. In our future work, we intend to delve deeper into exploring data quality concerns. This investigation will aim to address and understand any potential issues related to data noise, ensuring a more comprehensive evaluation of the datasets used in our methodology. This effort will enhance the reliability and robustness of our approach by minimizing the impact of any inaccuracies brought in during the phase of manual tagging phase.

6. Conclusion

With the rapid development of information technology, log data has become a crucial basis for monitoring and diagnosing system behavior. This paper proposes a novel anomaly detection method based on log data analysis called DualAttlog. This method integrates self-matching networks, pre-trained models (such as RoBERTa), and dual attention networks to capture deep semantic information in log sequences and effectively identify anomalies. The self-matching network explores interactions between words in logs, the pre-trained model provides higher-level semantic information, and the dual attention network integrates inconsistent and synthetic information. DualAttlog performs well on multiple benchmark datasets, outperforming other advanced models and demonstrating robustness to real-world data. This method has extensive potential and applicability in practical applications.

Although our research has made certain progress, there are inevitably limitations and shortcomings. Firstly, the current model primarily focuses on offline log analysis and has not been verified in actual online systems, which limits the evaluation and improvement of the model's effectiveness in practical applications. Secondly, the model's training and inference speed still need to be improved to meet the high-efficiency requirements of real-time log processing. These two points provide clear directions and optimization opportunities for future research.

Additionally, with the rapid development of large language model technology, how to efficiently apply this advanced technology to the field of log anomaly detection has become the focus of our next research. We expect that through in-depth research and exploration, we can integrate the powerful capabilities of large language models into log analysis, further enhancing the accuracy and efficiency of log anomaly detection, and providing more solid guarantees for system stability and security.

CRedit authorship contribution statement

Haitian Yang: Writing – review & editing, Writing – original draft, Methodology. **Degang Sun:** Visualization, Validation, Formal analysis. **Weiqing Huang:** Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no financial and personal relationships with other people or organizations that can inappropriately influence their work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

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